

Classification Of Magnetic Materials On The Basis of Magnetic Moment

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Introduction

- ▶ **Magnetic Materials** are those materials in which a state of magnetization can be induced. In other words, The materials which can be magnetized are known as Magnetic Materials.
- ▶ **Magnetic Moment** is a measure of the strength of a magnet. It is the product of strength of one of the poles and the distance between the two poles of a magnet.

Classification Of Magnetic Material

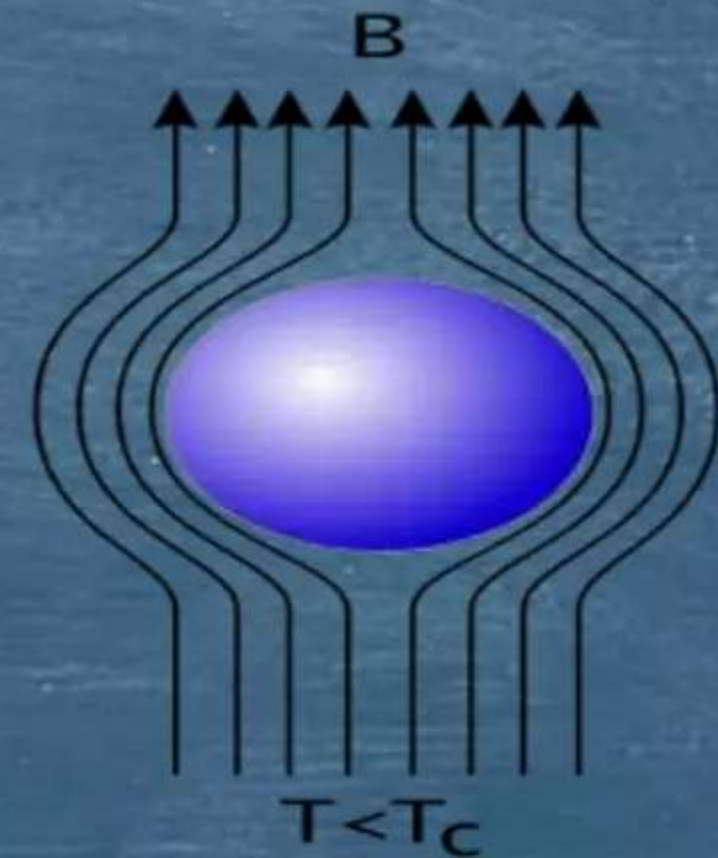
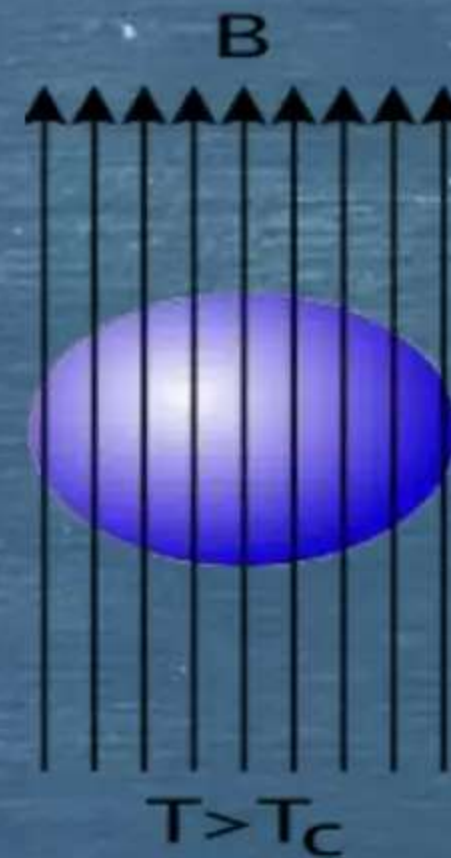
- ▶ Magnetic materials are classified into different categories based on their magnetic parameters. And also on the basis of effect of temperature and magnetic field on the magnetic properties.
- ▶ So, all materials are classified broadly into the following three categories
 - ▶ Diamagnetic Materials
 - ▶ Paramagnetic Materials
 - ▶ Ferromagnetic Materials
 - ▶ Antiferromagnetic Materials
 - ▶ Ferrimagnetic Materials



These are having very close structure to ferromagnetic materials but possess different magnetic effect.

Diamagnetic Materials

- ▶ **Diamagnetic** materials create an induced magnetic field in a direction opposite to an externally applied magnetic field.
- ▶ They are repelled by the applied magnetic field.
- ▶ The permanent dipoles are absent in Diamagnetic materials

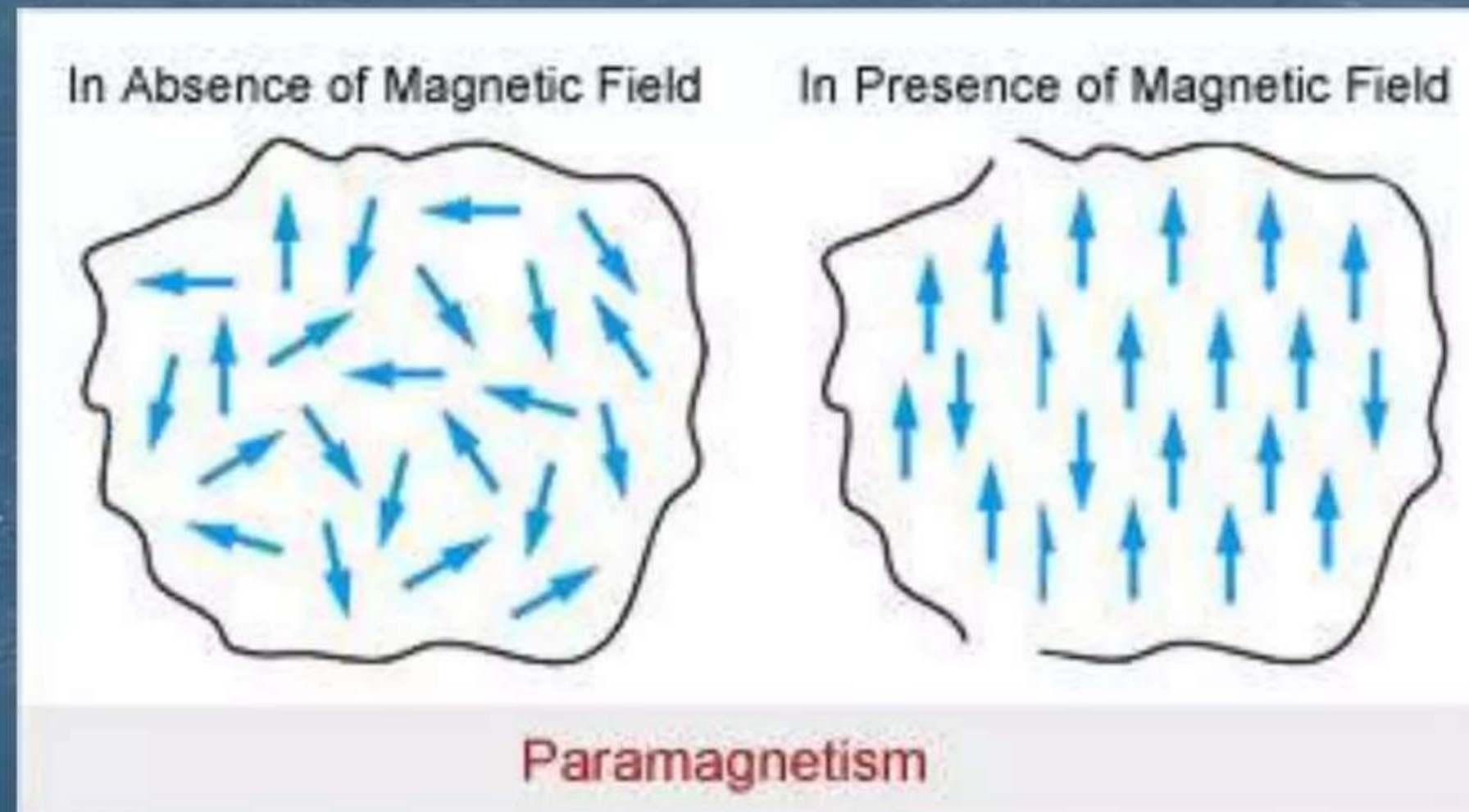


General Properties of Diamagnetic Materials

- ▶ Diamagnetic Materials experiences a repelling force when brought near the pole of a strong magnet.
- ▶ The magnetic susceptibility χ of these materials is always negative.
- ▶ The relative permeability μ_r is always less than one.
- ▶ In the absence of external magnetic field, The net magnetic dipole moment over each atom or molecule of a diamagnetic material is zero. This is due to pairing of electrons.
- ▶ Examples:-Bismuth,Copper,Lead,Zinc etc.

Paramagnetic Material

- ▶ Paramagnetic materials exhibit magnetism when the external magnetic field is applied. Paramagnetic materials lose magnetization in the absence of an externally applied magnetic field. These materials are weakly attracted towards magnetic field.

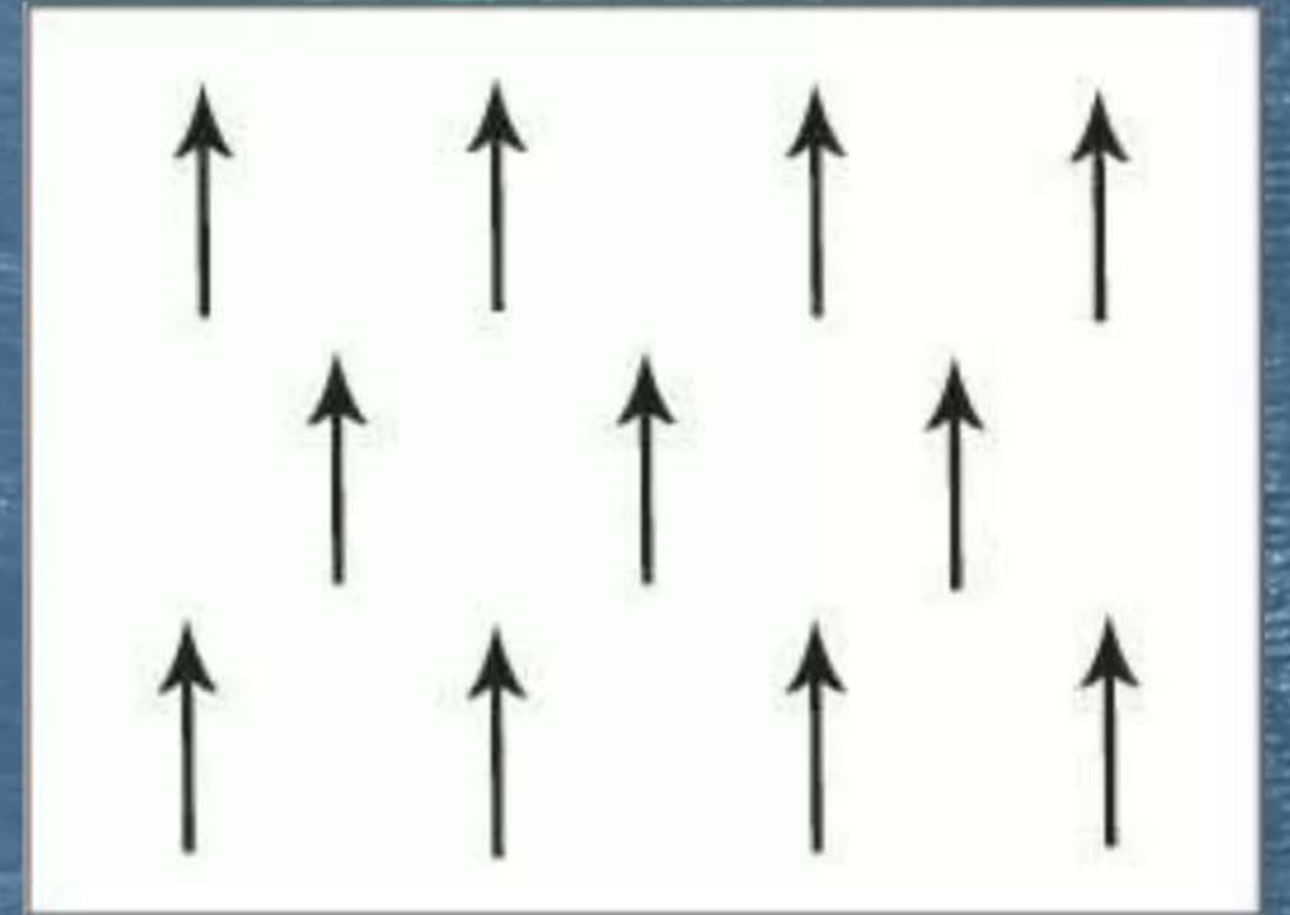


General Properties of Paramagnetic materials

- ▶ Paramagnetic materials experiences a feeble attractive force when brought near the pole of a magnet
- ▶ These materials possess some permanent dipole moment which arise due to some unpaired electrons.
- ▶ The magnetic susceptibility χ is small and +ve.
- ▶ Examples:-Platinum,Aluminium,Copper sulphate etc.

Ferromagnetic Material

- ▶ It is the phenomenon in which a material gets magnetized to a very large extent in the presence of an external field.
- ▶ The direction in which the material gets magnetized is the same as that of the external field.



General Properties of ferromagnetic materials

- ▶ Ferromagnetic materials experience a very strong attractive force when brought near the pole of a magnet.
- ▶ Permeability is very much greater than one.
- ▶ Susceptibility is +ve and high.
- ▶ Examples:-Fe,Co,Ni,MnAs etc.

Ferromagnetic Materials

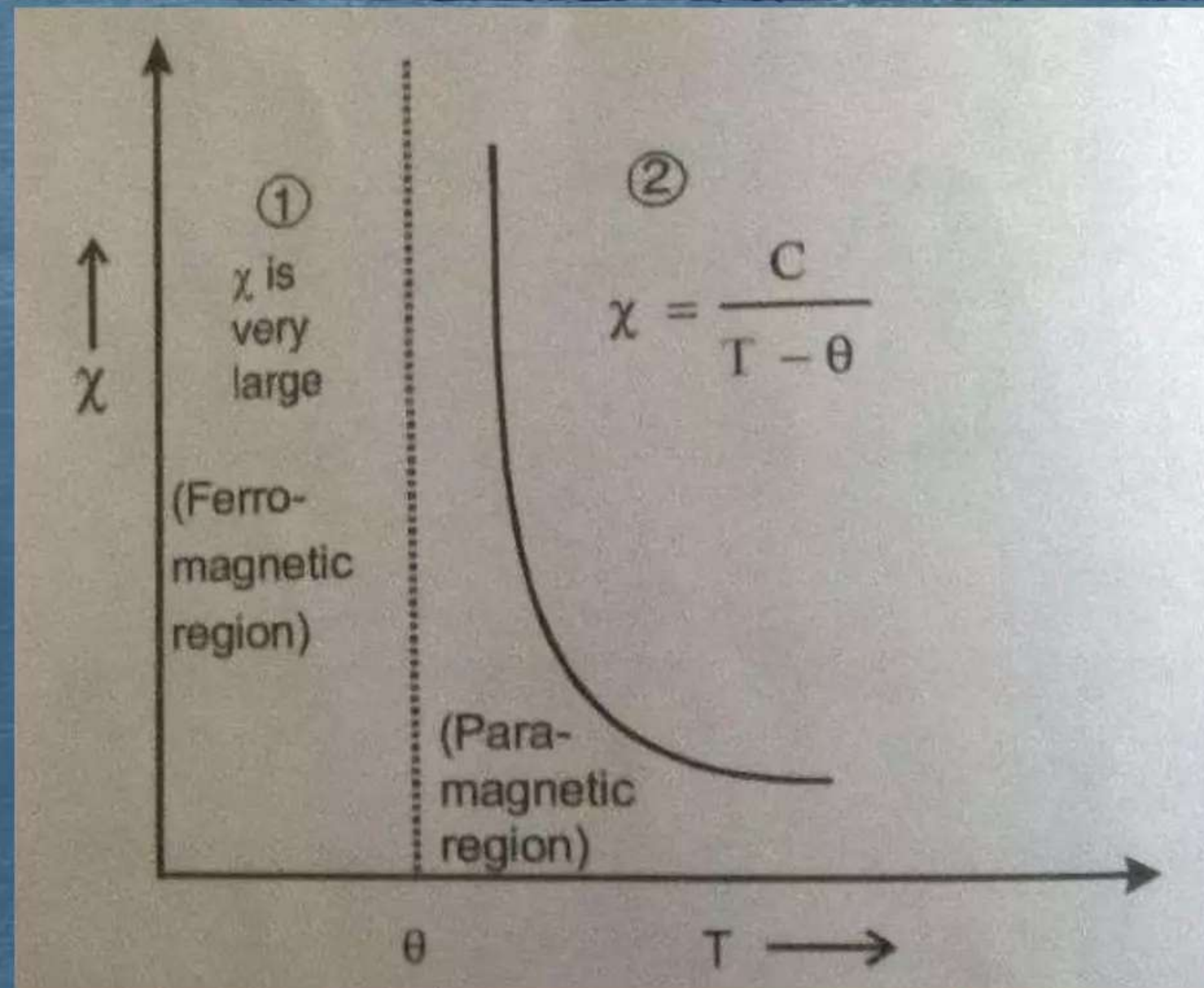
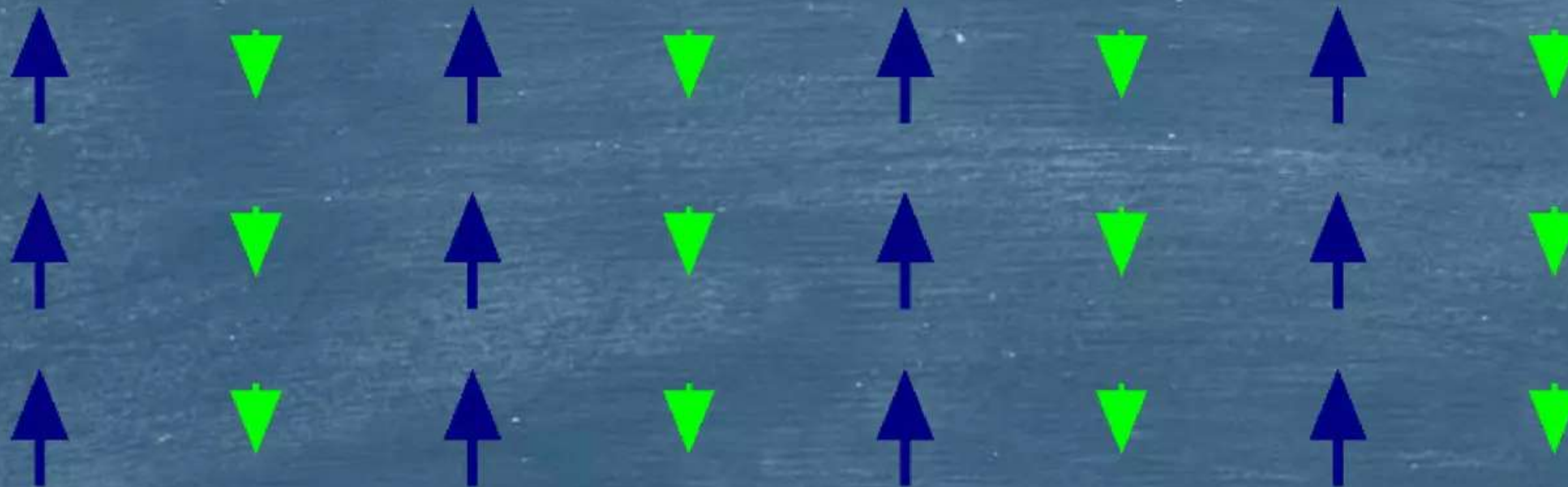


Figure. Variation Of susceptibility with temperature in a ferromagnetic material

Antiferromagnetic Material

- ▶ It is refer to a phenomenon in which the magnetic interaction between any two dipoles align themselves anti-parallel to each other.
- ▶ Since all dipoles are of equal magnitude , the net magnetization is zero.



General properties of Antiferromagnetic materials

- ▶ Like ferromagnetic materials antiferromagnetic materials also possess dipole moment due to spin of the electron.
- ▶ The opposite alignment of adjacent dipoles due to an exchange interaction.
- ▶ The susceptibility is very small and is +ve.

Antiferromagnetic materials

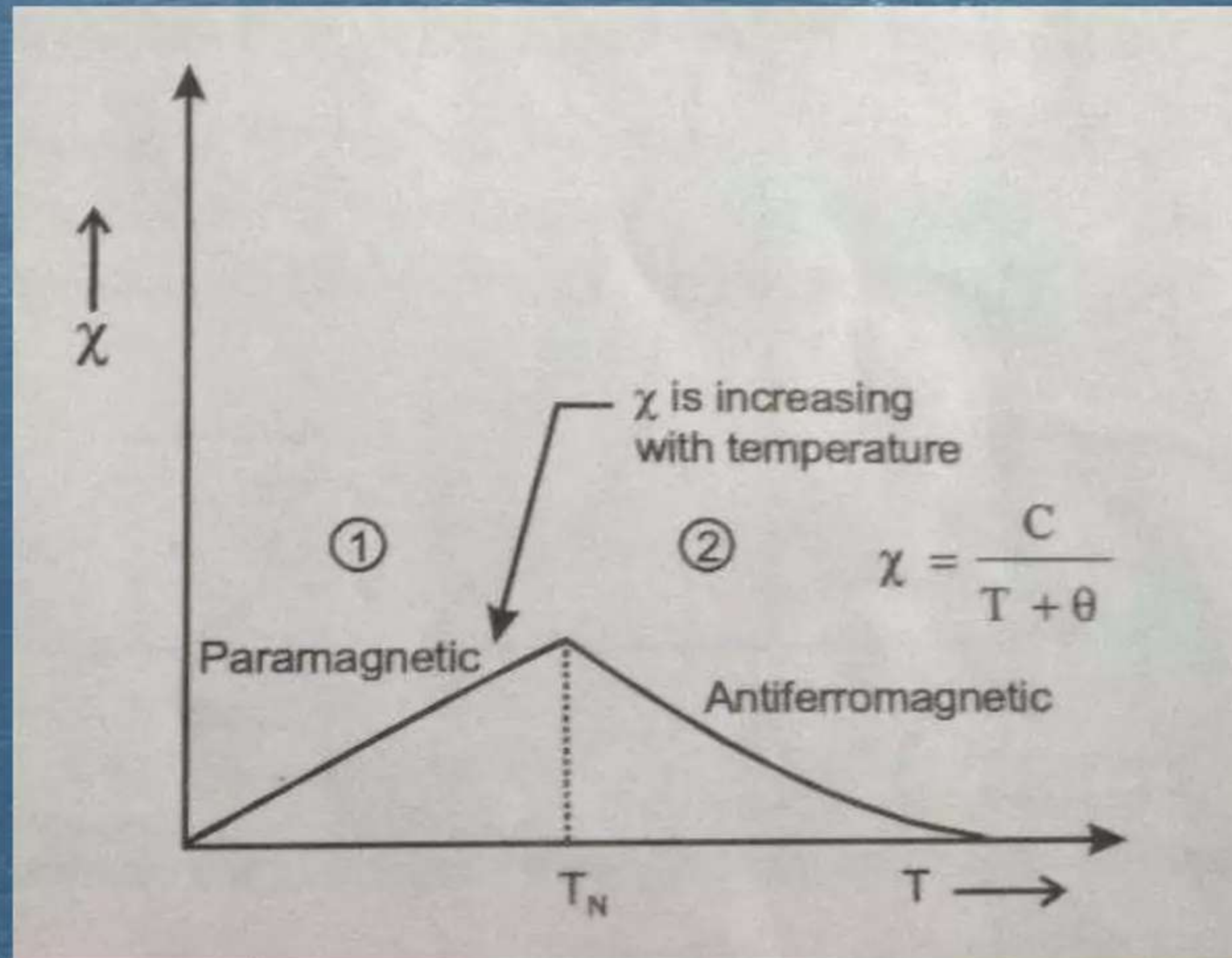


Figure. Variation of the susceptibility with temperature in antiferromagnetic material

Ferrimagnetic Material

- ▶ Ferrimagnetism is a phenomenon in which the magnetic interaction between any two dipoles align anti-parallel to each other.
- ▶ But since the magnitude of dipoles are not equal.
- ▶ The cancellation of magnetic moments become incomplete resulting in a net magnetization in the material.

General properties of Ferrimagnetic materials

- ▶ Ferrimagnetic materials possess magnetic dipoles moment due to the spin of the electron.
- ▶ A Ferrimagnetic material is composed of more state of different transition elements.
- ▶ The susceptibility is very Large and +ve.
- ▶ Examples:-Nickel,Ferrite and Ferrous ferrite.

Ferrimagnetic materials

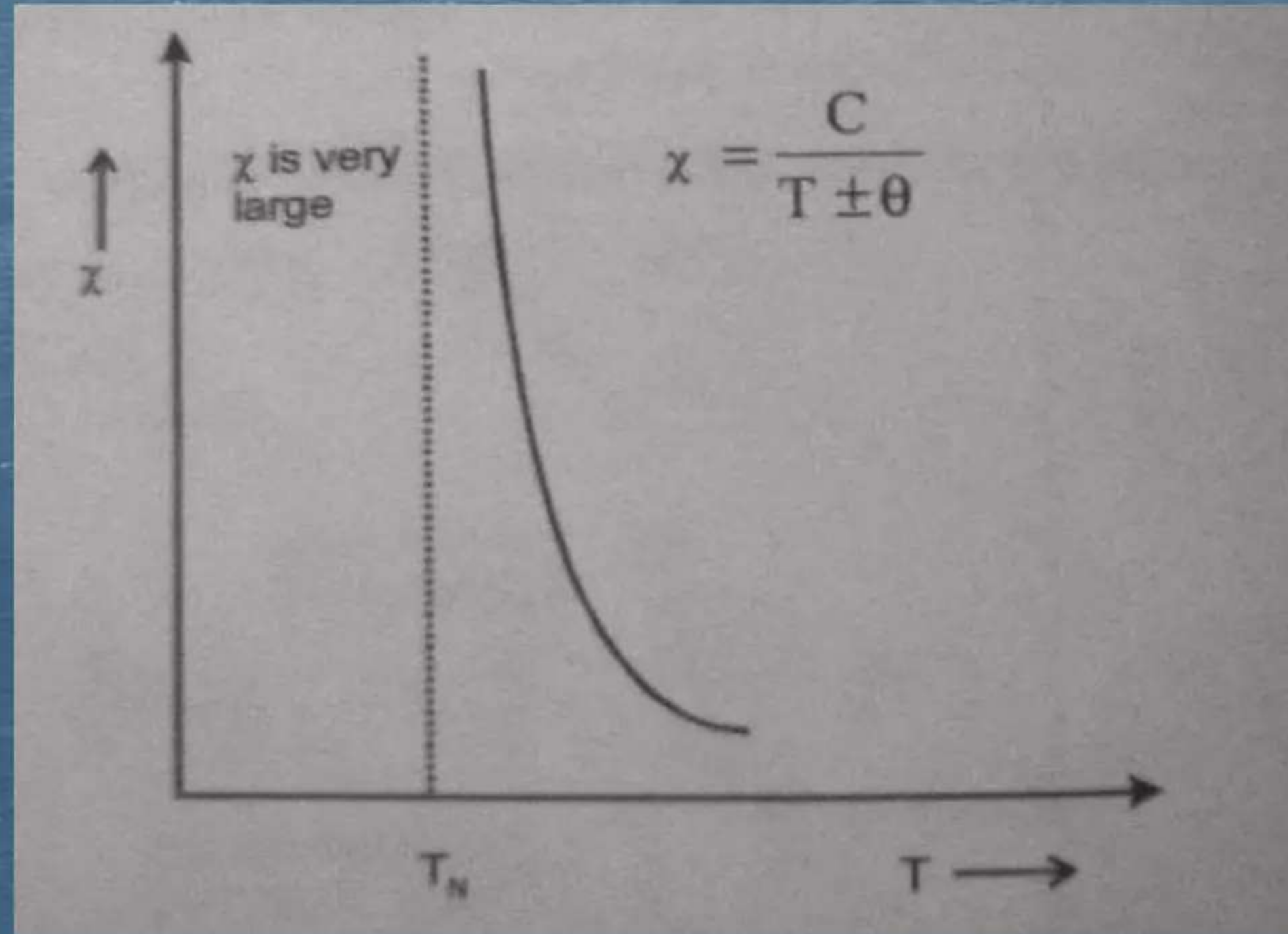
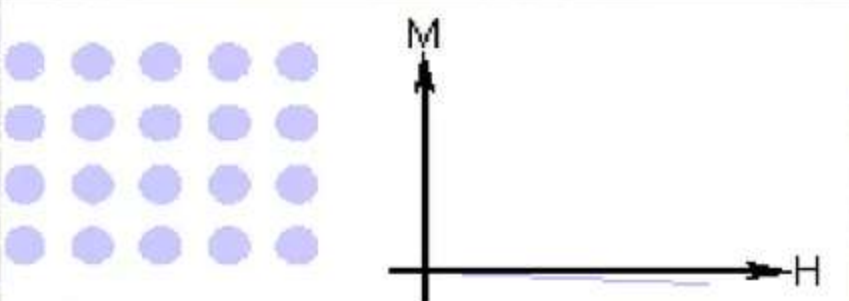
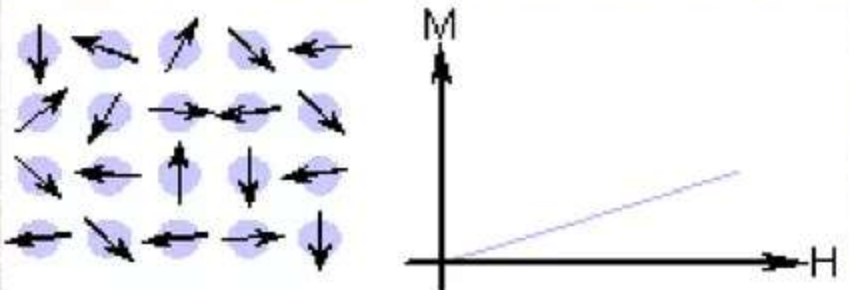
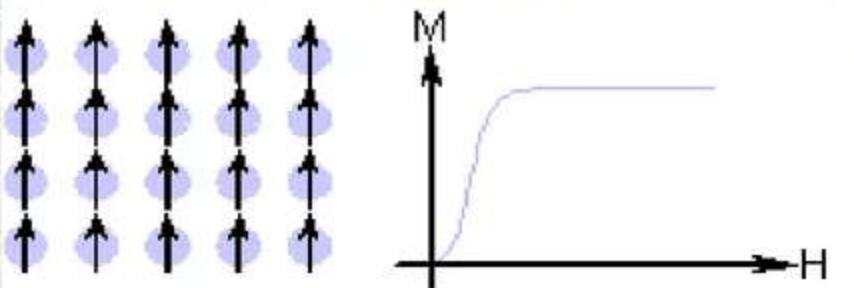
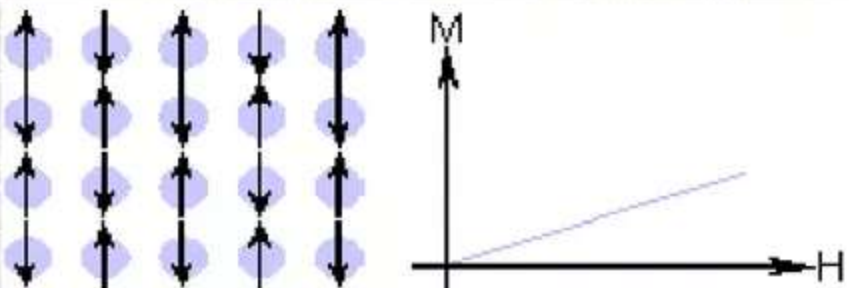
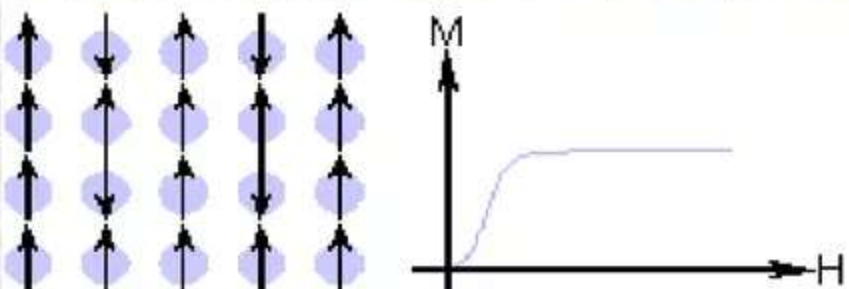


Figure. Variation of susceptibility with temperature in ferrimagnetic materials

Overview

Type of Magnetism	Susceptibility	Atomic / Magnetic Behaviour		Example / Susceptibility
Diamagnetism	Small & negative.	Atoms have no magnetic moment		Au Cu -2.74×10^{-6} -0.77×10^{-6}
Paramagnetism	Small & positive.	Atoms have randomly oriented magnetic moments		β -Sn Pt Mn 0.19×10^{-6} 21.04×10^{-6} 66.10×10^{-6}
Ferromagnetism	Large & positive, function of applied field, microstructure dependent.	Atoms have parallel aligned magnetic moments		Fe $\sim 100,000$
Antiferromagnetism	Small & positive.	Atoms have mixed parallel and anti-parallel aligned magnetic moments		Cr 3.6×10^{-6}
Ferrimagnetism	Large & positive, function of applied field, microstructure dependent	Atoms have anti-parallel aligned magnetic moments		Ba ferrite ~ 3

